

HAIG-SIMONS INCOME^{*}

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Abstract

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1 Theoretical Framework

Aggregate output Y_t is produced by a representative firm with production function $Y_t = F(K, L)$, where K is capital and L is labor. Firm profit equals $\Pi_t = F(K_t, L_t) - w_t L_t - \rho_t K_t$, where w_t is the wage rate, and ρ_t the rental rate of capital.¹ All assets are owned by the business sector, and business income (BI_t) includes both (implicit) capital rental income as well as profits: $BI_t \equiv \Pi_t + \rho_t K_t - \delta K_t$. Out of this income, firms either pay dividends d_t or retain the earnings $RE_t = BI_t - d_t$.

Capital evolves through the law of motion $K_{t+1} = (1 - \delta)K_t + I_t$. Investment is the sum of retained earnings (gross of depreciation) and net equity issuances, NE_t , which equals share issuances minus buybacks. We then have $I_t = RE_t + \delta K_t + NE_t$.

The market value of the firm (share price P_t times shares outstanding S_t) equals the value of its capital plus the present discounted value of future profits, denoted by V_t .

$$P_t S_t = K_t + E_t \left[\sum_{s=1}^{\infty} M_{t,t+s} \Pi_{t+s} \right] \equiv K_t + V_t, \quad (1)$$

where $M_{t,t+s}$ is the stochastic discount factor. Implicit in this equation is the Miller and Modigliani [1961] assumption that a dollar of retained earnings increases the value of the firm by a dollar.

We furthermore assume that equity issuances and buybacks are at the prevailing market price, $NE_t = P_t(S_t - S_{t-1})$. This ensures that changes in net equity have no effect on the price per share.²

There are two distinct types of income present in this economy.

Definition 1. (*Net of depreciation*) *National Income is equal to income from wages plus dividends plus retained earnings.*

$$Y_t^n = w_t L_t + d_t + RE_t \quad (2)$$

National income is income received from production. Definition 1 is in line with the BEA definition of national income, which, absent measurement errors, equals the net national product: $Y_t^n = Y_t - \delta K_t$.³

Definition 2. *Capital gains equal the real increase in the price per share of equity times the value of equity. Defining $r_t^p = P_t/P_{t-1} - 1$, $KG_t = P_{t-1}S_{t-1} \cdot r_t^p$.*

Capital gains are the increase in wealth caused by changes in asset prices. From Equation 1, the law of motion for capital, and the share repurchase equation, we

¹These are thus ‘pure’ or factorless profits, as in Karabarbounis and Neiman [2019].

²To see this, define share holder value as $W = PS$ and consider a deviation in net equity raised Δ , at a price per share of P . Then new firm value is $W' = W + \Delta$, and new shares outstanding is $S' = S + \Delta/P$. The price per share is then $W'/S' = (PS + \Delta)/(S + \Delta/P) = P$.

³See the BEA Handbook, Fox and McCully [2009].

can decompose gains into corporate retained earnings and changes in the value of future discounted profits:

$$KG_t = \Delta V_t + RE_t. \quad (3)$$

The BEA definition of National Income explicitly excludes capital gains, which are not earned in production. However, as shown in Equation 3, capital gains are indirectly present through retained earnings. To isolate capital gains from other income flows, we define ordinary factor income as national income minus the retained earnings of corporations: $Y_t^o = Y_t^n - RE_t$, and *pure* capital gains as capital gains minus retained earnings, $KG_t^p = KG_t - RE_t$.

To measure aggregate capital gains in the data, we use data from the Financial Accounts of the Federal Reserve, which collects information on the balance sheet of U.S. household sector, including capital gains which correspond to Definition 2. We convert nominal to real gains by subtracting net national product inflation, and separate total from pure capital gains by subtracting the BEA series for corporate savings.

Figure 2 displays the results. Capital gains are large, especially in the post-1980 period, where they average 14% of national income. Pure capital gains are the majority, averaging 10% of national income, while retained earnings, the component present in national income, averaged 4%. The large magnitude and persistence of capital gains suggests that analyses that do not include this income source provide an incomplete picture of inequality in recent times.

The growth in the economic importance of capital gains motivates a comprehensive income definition that includes changes in asset prices as well as standard income components.

Definition 3. *Haig-Simons income [Haig, 1921, Simons, 1938] equals ordinary national income plus capital gains: $HS_t = Y_t^o + KG_t$.*

Compared with other income definitions, Haig-Simons has several desirable properties.

Remark 1. *The change in household wealth between two periods is equal to Haig-Simons income (on a post-tax post-transfer basis) minus consumption: $\Delta W_t = W_t - W_{t-1} = HS_t - c_t$.*

To see this, we introduce a household sector that holds all assets, with flow budget constraint given by

$$c_t + P_t S_t = S_{t-1}(P_t + d_t) + w_t L_t \quad (4)$$

Defining wealth as $W_t = P_t S_t$, $\Delta W_t = (Y_t^n - c_t) + KG_t^p$. Haig-Simons income can thus more accurately account for changes in wealth compared to national income and national saving, which do not include revaluations.

Capital gains are also closely related to consumer welfare.

Remark 2. *From the household’s budget constraint, capital gains and other capital income sources are interchangeable: a dollar of gains expands the agent’s feasible consumption set identically to a dollar of interest or dividends.*

Moreover, holding expected returns constant, a dollar of capital gains increases welfare by the marginal utility of consumption. For this reason, when studying relative inequality, Haig-Simons is preferred to other income concepts since it better distinguishes welfare not only between asset holders and non-asset holders, but between investors who hold different types of assets. Consider two investors with identical wealth invested in assets earning the same total return. Investor A holds a bond paying a 5% coupon; Investor B holds a growth stock that pays no dividend but appreciates 5% per year. Under national income, Investor A earns \$5 per \$100 invested, while Investor B earns zero — despite being economically identical. More broadly, any income measure that excludes capital gains will systematically understate the income of growth stock holders relative to dividend stock holders, equity holders relative to bond holders, and business founders relative to salaried executives.

What are the sources of aggregate capital gains, and why have they grown over time?

Remark 3. *From Equations 1 and 3 Capital gains may be driven by (i) changes in firm profits (ii) changes in retained earnings (iii) changes in discount rates.*

In the U.S. context, there is substantial evidence that all three channels have played a role in the rise in capital gains. For (i), Grullon et al. [2019], Autor et al. [2020] document a secular increase in concentration and market power, which in turn has lead to increases in earnings and profits [Barkai, 2020, Eeckhout, 2024].

For (ii), although official measures of retained earnings have remained flat at around 4% of national income⁴, these figures do not include a large stock of *intangible investment*, which are treated as expenses (rather than investments) on income statements, and thus represent unmeasured retained earnings.⁵ Corrado et al. [2009], Crouzet et al. [2022], and Peters and Taylor [2017] document a secular increase in intangible investment and capital in the U.S.

Outside of these secular shifts, standard trend-growth can also lead to capital gains. Under a balanced growth path in which all macroeconomic variables grow at rate g , from Equation 1 the ratio of capital gains to GDP will be $g \cdot \frac{P}{Y}$. Plugging in the value of public and private equities held by US households for P and a real GDP growth rate of 2% yields a back-of-the envelope estimation of 5% of national income in capital gains per year.

For (iii), there has been a decline real interest rates since the 1980s, both in the United States [Eggertsson et al., 2019] and globally [Rachel and Summers,

⁴Globally, Chen et al. [2017] shows that corporate saving has trended upwards since the 1980s.

⁵Although the BEA has revised its treatment of intangible investment to include additional categories such as software, it still excludes firm-specific human capital, organization capital, brand equity, and advertising.

2019]. Changes in discount rates complicate the interpretation of Haig-Simons income because in addition to the wealth effect of higher asset prices, future returns are also affected. We discuss the impact of lower interest rates (and more generally, changes in discount rates) in Section J.

Remark 4. *Changes in firm payout policy can affect the level of total capital gains KG_t , but not pure gains KG_t^p .*

Ceteris paribus, a change in stock buybacks holding dividends fixed has no effect on capital gains, since under our assumptions a fair-value repurchase leaves the price per share unchanged. If the increase in buybacks is instead matched by a decrease in dividends, however, aggregate firm value is unchanged while shares outstanding fall, leading to an increase in KG_t . Since pure gains net out retained earnings, KG_t^p is unchanged. For this reason, the shift in payout policy towards repurchases [Skinner, 2008] is not responsible for the trends in pure gains documented in Figure

Remark 5. *The Haig-Simons concept does not double count dividend income.*

[Canberra, 2001, Splinter, 2025] argue that Haig-Simons income double counts dividends, since (according to the argument) higher expected future dividend growth increase Haig-Simons income once when stock prices increase, and again when the dividends are paid out. This analysis, however, overlooks the fact that paying out dividends reduces the market value of the share approximately one-to-one with payouts [Allen and Michaely, 2003]; this is exactly the reason why stock prices fall on ex-dividend dates. The fact that dividend payments subtract from capital gains ensure there is no double counting of income.⁶

References

- Franklin Allen and Roni Michaely. Payout policy. *Handbook of the Economics of Finance*, 1:337–429, 2003.
- David Autor, David Dorn, Lawrence F Katz, Christina Patterson, and John Van Reenen. The fall of the labor share and the rise of superstar firms. *The Quarterly journal of economics*, 135(2):645–709, 2020.
- Simcha Barkai. Declining labor and capital shares. *The Journal of Finance*, 75(5):2421–2463, 2020.
- Canberra. Final report and recommendations. Technical report, The Canberra Group, Ottawa, 2001.
- Peter Chen, Loukas Karabarbounis, and Brent Neiman. The global rise of corporate saving. *Journal of monetary economics*, 89:1–19, 2017.

⁶A similar argument to this was made in Larrimore et al. [2021].

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- Carol Corrado, Charles Hulten, and Daniel Sichel. Intangible capital and us economic growth. *Review of income and wealth*, 55(3):661–685, 2009.
- Nicolas Crouzet, Janice C Eberly, Andrea L Eisfeldt, and Dimitris Papanikolaou. The economics of intangible capital. *Journal of Economic Perspectives*, 36(3): 29–52, 2022.
- Jan Eeckhout. The value and profits of firms. Technical report, CEPR Discussion Papers, 2024.
- Gauti B Eggertsson, Neil R Mehrotra, and Jacob A Robbins. A model of secular stagnation: Theory and quantitative evaluation. *American Economic Journal: Macroeconomics*, 11(1):1–48, 2019.
- Douglas R Fox and Clinton P McCully. Concepts and methods of the us national income and product accounts. *NIPA Handbook*, 2009.
- Gustavo Grullon, Yelena Larkin, and Roni Michaely. Are us industries becoming more concentrated? *Review of Finance*, 23(4):697–743, 2019.
- Robert M Haig. The concept of income-economic and legal aspects. *The federal income tax*, 1(7), 1921.
- Loukas Karabarbounis and Brent Neiman. Accounting for factorless income. *NBER Macroeconomics Annual*, 33(1):167–228, 2019.
- Jeff Larrimore, Richard V Burkhauser, Gerald Auten, and Philip Armour. Recent trends in us income distributions in tax record data using more comprehensive measures of income including real accrued capital gains. *Journal of Political Economy*, 129(5):1319–1360, 2021.
- Merton H Miller and Franco Modigliani. Dividend policy, growth, and the valuation of shares. *the Journal of Business*, 34(4):411–433, 1961.
- Ryan H Peters and Lucian A Taylor. Intangible capital and the investment-q relation. *Journal of Financial Economics*, 123(2):251–272, 2017.
- Łukasz Rachel and Lawrence H Summers. On secular stagnation in the industrialized world. Technical report, National Bureau of Economic Research, 2019.
- Henry C Simons. *Personal Income Taxation: The Definition of Income as a Problem of Fiscal Policy*. University of Chicago Press, Chicago, 1938.
- Douglas J Skinner. The evolving relation between earnings, dividends, and stock repurchases. *Journal of financial economics*, 87(3):582–609, 2008.
- David Splinter. The distribution of capital gains in the united states: A comment. 2025.